

Numerical Analysis PhD Exam, May 1999

Do any 8 of the following 10 problems.

1. This problem has six parts.

- (1) Give a careful statement and proof of the singular value decomposition.
- (2) Give a careful statement of the QR factorization theorem.
- (3) Indicate how the Householder transformations are used in the proof of the QR factorization theorem.
- (4) Describe the QR Iteration Algorithm with shift μ .
- (5) Describe how the QR Iteration is used to compute the SVD .
- (6) How can the singular values be used to compute the condition number of a matrix.

2. Give a careful proof of the following Theorem: If $p \geq 1$,

1. $\mathbf{A} \in \mathbb{R}^{n \times n}$ is invertible, 2. $\mathbf{b}$ and $\Delta \mathbf{b}$ are vectors in $\mathbb{R}^n$, and 3. $\mathbf{x}$ and $\Delta \mathbf{x}$ are vectors in $\mathbb{R}^n$ and are solutions to $\mathbf{A}\mathbf{x} = \mathbf{b}$, and $\mathbf{A}\Delta \mathbf{x} = \Delta \mathbf{b}$, respectively,

then

$$\frac{\|\Delta \mathbf{x}\|_p}{\|\mathbf{x}\|_p} \leq \kappa_p(\mathbf{A}) \frac{\|\Delta \mathbf{b}\|_p}{\|\mathbf{b}\|_p}.$$

3. This problem has three parts.

- (1) Describe how Jacobi rotations can be used to diagonalize a real symmetric matrix.
- (2) Give the error estimate for a single Jacobi rotation.
- (3) Give the error estimates for repeated Jacobi rotations.

4. This problem has two parts.

- (1) Write out the system of linear equations to be solved to find the periodic spline interpolation for the given data points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$, where $x_0 < x_1 < \dots < x_n$ and $y_n = y_0$.
- (2) What method would you recommend for solving the system of linear equations. Explain!!

5. This problem has three parts.

- (1) Give a careful definition of the Householder transformation.
- (2) Explain why the Householder transformation is preferred over the Gram–Schmidt orthogonalization process for creating an orthonormal basis. (Give an example if necessary.)
- (3) Show how Householder transformations can be used to bidiagonalize any real matrix.

6. This problem has four parts.

- (1) Give a careful statement of the Pythagorean Theorem property for clamped cubic splines.
- (2) Give a careful statement of the integral minimization property for clamped cubic splines.
- (3) If $s_n(x)$ denotes the clamped cubic spline interpolant for a function $f(x)$ at the knots $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$, then give a precise statement for the error between $s_n(x)$ and $f(x)$.

(4) Prove the error formula given in step (3) above.

7. This problem has three parts.

- (1) If $p_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$, then show that $\int_{-1}^1 x^m p_n(x) dx = 0$ for all $m < n$. (Hint: Think small values of m and n , induction, and parts.)
- (2) What can you conclude about the relationship between the polynomials $p_n(x)$ and the Legendre polynomials?
- (3) Explain the method of Gauss quadrature for the numerical approximation of the integral of a function.

8. This problem has three parts.

- (1) Discuss the ideas behind the linear shooting method for approximating the solution of a linear two point boundary value problem. In particular, explain how the Runge–Kutta method can be used in the approximation.
- (2) Describe how the Galerkin method can be used to solve a 2 point boundary value problem on the interval $[0, 1]$.
- (3) Explain why splines have an advantage over $\sin(kx)$ as choices of test functions for the Galerkin method.

9. This problem has three parts.

- (1) Give a careful statement of the Contraction Mapping Theorem.
- (2) Outline a proof of the Contraction Mapping Theorem.
- (3) Indicate how Aitken's method can be used to accelerate convergence.

10. This problem has two parts.

- (1) Give a careful statement of the error formula for the trapezoidal rule for approximating an integral.
- (2) Give a careful proof of the error formula for the trapezoidal rule.